# Flag Football: a data product, end to end

SYNTHETIC DATA

Three business goals, turned into five data products that are owned, measured and safe for kids.

1 · WHAT WOULD YOU BUILD FOR THESE GOALS?

## Each goal has one product that moves it directly.

Three goals led to five products.

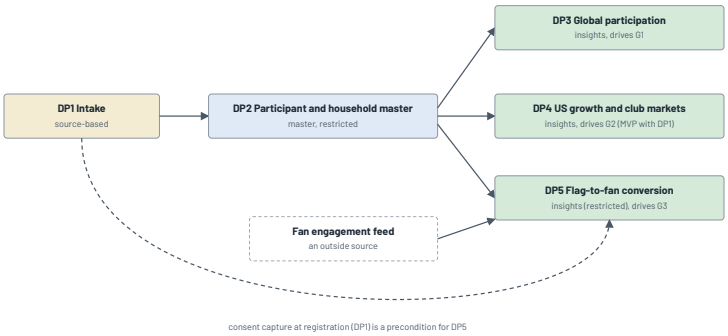
| Goal | DP1 Intake | DP2 Master | DP3 Global | DP4 US growth | DP5 Fan conversion |
|---|---|---|---|---|---|
| **G1** Grow the NFL globally | Utility | Enabler | Driver | | |
| **G2** Grow flag in the US as a safe, low-cost way into football | Utility | Enabler | | Driver | |
| **G3** Turn families into fans | | Enabler | | | Driver |

Driver: moves the goal directly. Enabler: makes other products possible. Utility: keeps things running.

2 · WHAT WOULD YOU BUILD FOR THESE GOALS? (CONTINUED)

## Intake feeds the master, and the master feeds the three insight products.

Conversion also needs the fan feed and guardian consent.




3 · WHO ARE THE PRODUCTS FOR?

## Six personas, and what we'd ask them.

There are no real users yet, so these are assumptions to test.

### International growth lead

Serves G1

Decide which countries to invest in

**We would ask:** Which decisions did you make last year without data?

### NFL FLAG domestic operations lead

Serves G2

Know where flag is growing and where leagues are struggling

**We would ask:** Which numbers do you distrust today, and why?

### Club youth and community manager

Serves G2, G3

Grow flag in our market and connect families to the club

**We would ask:** What would you do differently with a market view?

### Fan and club marketing analyst

Serves G3

Show that flag creates fans, and reach families who agreed to be reached

**We would ask:** What evidence would convince finance that flag drives fandom?

### Privacy and compliance lead

Serves G3 (guardrail)

Make sure children's data is never misused

**We would ask:** What would make you comfortable approving club access?

### League operator (volunteer)

Serves G2 (supplier)

Run the league and send the data that is asked for

**We would ask:** What is the most you would fill in at registration?

4 · WHAT IS EACH PRODUCT?

## Each product has a label: owner, version, live quality grade and rules for use.

Open one to see who uses it and what it must never do.

| Product | Type | Status | Version | Quality | Owner | Rule |
|---|---|---|---|---|---|---|
| DP1 Intake | Source-based | MVP · building | v2.0.0 | Grade C | Data Product Manager, NFL FLAG Insights (fictional) | Never: re-identification of a child from an intake file |
| DP2 Master | Master | Proposed | v0.10.0 | Identity error under 1% | Data Product Manager, NFL FLAG Insights (fictional) | Never: marketing access to the master |
| DP3 Global | Insights | Proposed | v0.9.0 | Grade C | Data Product Manager, NFL FLAG Insights (fictional) | Never: publishing a country count below the small-cell size |
| DP4 US growth | Insights | MVP · building | v1.1.0 | Grades vary by market | Data Product Manager, NFL FLAG Insights (fictional) | Never: individual targeting |
| DP5 Fan conversion | Insights | Proposed | v0.1.0 | Grade D | Data Product Manager, NFL FLAG Insights (fictional) | Never: child-level targeting |

Quality grade: A is 95 or more, B is 85, C is 70, D is below 70 (score out of 100). Identity error is how far identity matching alone could move the count. Grades vary by market: each market is graded on its own.

5 · WHAT DO WE BUILD FIRST, AND WHY?

## Intake and US growth first. Conversion waits: high value, data not ready.

Each product is scored on value, feasibility, readiness and privacy risk.

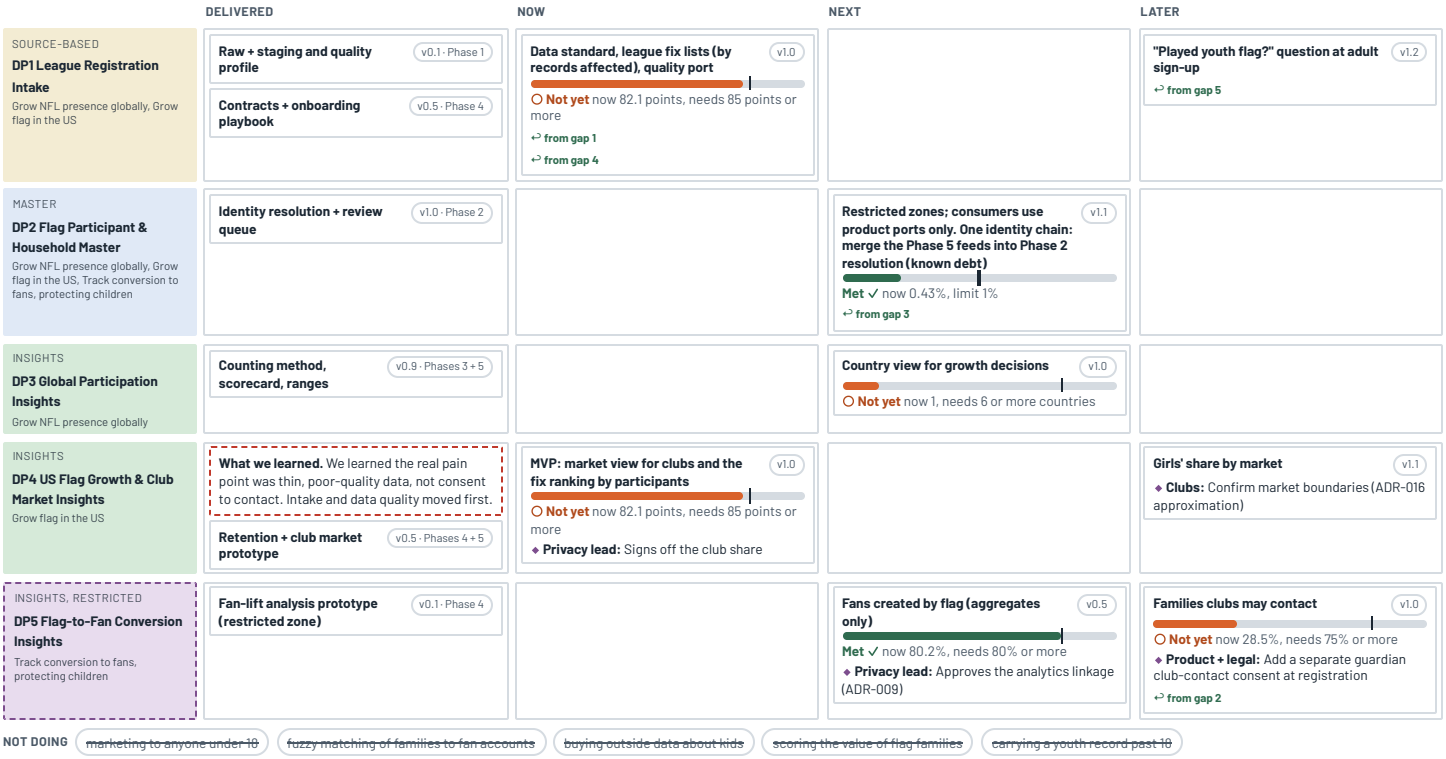
| Product | Value | Feasibility | Readiness | Privacy risk | Total | Reading |
|---|---|---|---|---|---|---|
| DP1 Intake | 4 | 5 | 3 | 1 | **11** | Build first |
| DP4 US growth | 5 | 4 | 3 | 2 | **10** | Build first |
| DP2 Master | 4 | 4 | 4 | 4 | **8** | Next |
| DP3 Global | 4 | 3 | 2 | 2 | **7** | High value, low readiness: sequenced, not deprioritized |
| DP5 Fan conversion | 5 | 2 | 1 | 5 | **3** | High value, low readiness: sequenced, not deprioritized |

Scores 1 to 5. Total = value + feasibility + readiness - privacy risk.

6 · WHAT HAS TO BE TRUE BEFORE EACH PRODUCT MOVES?

## 2 of 6 release gates met. The MVP waits on the US data score (82 → 85).

No dates: a product moves when its gate is met and the right person signs off.




No dates: products move when their data is good enough and the right person has signed off. Bars show where a gate stands; the tick is the target. ◆ marks a decision. Targets are set for a synthetic dataset.

7 · HOW DO WE PROTECT KIDS?

## Minors are guardian-only. We count families, never kids.

9% of families with a child under 13 agreed to club contact.

| Under 13 | 13–15 | 16–17 | 18+ |
|---|---|---|---|
| Guardian | Guardian | Guardian | Self, fresh consent |
| Counted in totals · club contact only via guardian consent | Counted in totals · club contact only via guardian consent | Counted in totals · club contact only via guardian consent | Own fan record, only with fresh consent |
| **9%** agreed to club contact: 520 of 5,790 families | **9%** agreed to club contact: 190 of 2,230 families | **8%** agreed to club contact: 190 of 2,250 families | **5%** agreed to club contact: 60 of 1,350 families |

⊘ ~~Marketing to under 18s~~ · ⊘ ~~Buying data about kids~~ · ⊘ ~~Fuzzy family matching~~ · ⊘ ~~Scoring families~~ · ⊘ ~~Youth records past 18~~

8 · IS IT WORKING?

## Fans created by flag: about 1,300 (900–1,600).

One row per goal, then the health of each product.

G1
**7,400**

7,400 players in 8 countries. Only 1 reports at grade B or better.

Not yet: now 1, needs 6 or more countries

**So what.** We can count globally, but can't compare countries fairly yet.

G2
**47%**

Girls 47% · 58% play again · 14 of 76 leagues sent no file.

Not yet: now 82.1 points, needs 85 points or more

**So what.** Growth looks real, but missing leagues mean we undercount it.

G3
**1,300**

About 1,300 new fans (900–1,600) · engaged families +20% · 36% of families linked.

Not yet: now 28.5%, needs 75% or more

**So what.** Flag does create fans. With only 36% of families linked, this is a floor, not the full number.

2025, synthetic

| Product | Quality | Tests | Privacy checks | Status |
|---|---|---|---|---|
| DP1 Intake | Grade C | ✓ | ✓ | **Watch** Quality: C |
| DP2 Master | Identity error under 1% | ✓ | ✓ | **Watch** Age-rule rows still unverified: 23 |
| DP3 Global | Grade C | ✓ | ✓ | **Watch** Quality: C |
| DP4 US growth | Grades vary by market | ✓ | ✓ | **Watch** Quality: A: 12, B: 4, C: 5, D: 5 |
| DP5 Fan conversion | Grade D | ✓ | ✓ | **Watch** Quality: D |

9 · WHERE ARE THE GAPS, AND WHAT DID THEY CHANGE?

## 7 gaps found. 5 of them changed the roadmap.

Each gap shows its evidence and the card it moved.

| Gap | Evidence | Product | What it changed | Roadmap |
|---|---|---|---|---|
| US data score is below target | now 82.1 points, needs 85 points or more | DP4 | Blocks the DP1 and DP4 release; intake and quality went first (the pivot) | Changed roadmap |
| Consent fully known for too few families | now 28.5%, needs 75% or more | DP5 | Club contact moved to Later; guardian consent added at sign-up as a decision | Changed roadmap |
| A second identity chain was found | planned for v1.1 | DP2 | Added the "one identity chain" card | Changed roadmap |
| Leagues that sent no file | 14 of 76 leagues | DP1 | The send-your-file fix ranks first on every fix list | Changed roadmap |
| Youth-to-adult link cannot be measured | Fresh consent at 18 and no carry-over: a youth record is not linked to the adult by design | DP5 | Added the played-youth-flag question at adult sign-up | Changed roadmap |
| Age rules not yet verified | 23 of 28 rules | DP1 | No roadmap change; the strictest rule applies | Watch |
| Too few countries at grade B | now 1, needs 6 or more countries | DP3 | No roadmap change; it is the gate for the country view | Watch |

10 · WHAT DID WE LEARN, AND WHAT'S NEXT?

## We started with consent. The real problem was thin, messy data.

| | What | So what |
|---|---|---|
| Pivot | Started with consent to contact; the real problem was thin, messy data. | We fixed supply first. A better model on bad data is still wrong. |
| Known debt | A second identity chain; merge planned (DP2 v1.1). | We track shortcuts openly and pay them down: boundary breaks went from 23 to 0. |
| Next decision | The privacy lead approves the fan analysis; product and legal add guardian consent at sign-up. | It unlocks DP5: the first time clubs can reach families, with consent. |

All figures synthetic, built to test the method · Full detail: Data product tab → · Methodology 2.0